

Training Cognitive Search Heuristics via Multilayer Perceptron (MLP) LoRA Adapters for Resource-Constrained Agentic Workflows

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Abstract

Context window expansion has emerged as the default brute-force solution for enabling Large Language Models (LLMs) to perform complex log analysis and system diagnostics. However, the quadratic activation memory complexity $\mathcal{O}(N^2)$ of standard Self-Attention mechanisms creates severe hardware constraints, frequently resulting in Out-Of-Memory (OOM) failures or extreme compute latency on commodity hardware. In this paper, we propose and evaluate an alternative paradigm: instead of expanding the active context window, we target the Multilayer Perceptron (MLP) feed-forward layers of a small model (Qwen2.5-3B-Instruct) with Quantized Low-Rank Adaptation (QLoRA) to encode **cognitive search heuristics** directly into the model’s weights. This teaches the model *how* to plan and search its context dynamically rather than memorizing raw text. We evaluate our approach in two phases. In Phase 1, we show that MLP-LoRA achieves a **39% generation speedup** and **100% formatting stability** on a 1,200-token server log haystack search compared to the baseline. In Phase 2, we benchmark our Heuristic 3B Agent on a complex 13,546-token multi-file diagnostics task against a baseline brute-force 3B model and a larger 7B model. While baseline brute-force ingestion immediately crashes due to CUDA OOM on a single 11GB RTX 2080 Ti, and host CPU brute-forcing takes **46.89 minutes (2,813.12s)**, our Heuristic 3B Agent successfully maps the compromise chain in **39.10 seconds**—representing a **72x speedup** while maintaining a peak active context under 2,741 tokens and a peak VRAM footprint under 1.5GB.

1 Introduction

Modern agentic workflows demand that Large Language Models (LLMs) act as autonomous diagnosticians, navigating massive streams of system logs, database transactions, and security audits to identify vulnerabilities [1]. Standard state-of-the-art systems rely on large context windows (e.g., 128k to 1M tokens) to ingest all system state at once. However, this brute-force ingestion suffers from three fatal flaws:

1. **Quadratic Activation Scale:** The memory required by Scaled Dot-Product Attention (SDPA) scales quadratically $\mathcal{O}(N^2)$ with prompt length N , leading to instant CUDA OOM on mainstream consumer GPUs [2].
2. **Distraction & ”Lost-in-the-Middle” Effect:** Ingesting huge dumps of noisy text degrades search success as the model struggles to locate small needles in vast haystacks [3].
3. **Execution Latency:** Prefilling large sequences is computationally expensive, making real-time interactive agents slow and costly.

Current neural interpretability research (e.g., ROME [4], MEMIT [5], and FFN key-value theories [6,7]) demonstrates that

while attention heads primarily dictate *where* the model shifts its focus in a prompt, factual associations, non-linear feed-forward behaviors, and policy execution are stored predominantly in the **Multilayer Perceptron (MLP)** layers.

We leverage this insight to show that by targeting the MLP layers (gate_proj, up_proj, down_proj) with LoRA [8], we can train a model’s *cognitive intuition* on how to search. The resulting agent maps out system directories and dynamically reads segments only as-needed, keeping the active context small, fast, and VRAM-efficient.

2 Related Work

Factual and Behavioral Storage in MLPs: Meng et al. [4,5] established that MLP feed-forward networks act as key-value databases storing factual associations. Geva et al. [6,7] further proved that FFN layers operate as key-value memories where the first linear layer acts as keys representing input patterns, and the second acts as values producing next-token distribution modifications. We extend this by showing that MLPs can also store *cognitive planning associations*—linking specific token signatures (e.g., directory structures or log anomalies) with agent

search behaviors.

Quantized Low-Rank Adaptation (QLoRA): Dettmers et al. [9] proved that loading models in 4-bit NormalFloat (nf4) and training high-precision LoRA adapters preserves full fine-tuning performance. We adapt QLoRA for native bfloat16 compute to achieve gradient convergence without GradScaler overhead on Turing architecture GPUs, which lack native hardware-accelerated bfloat16 scaling support.

Agentic Planning vs. Brute Ingestion: Rather than expanding the physical context window (e.g., FlashAttention-2 [10], vLLM [11]), we optimize the *agentic search strategy* itself, treating context management as a sequential decision-making process. This builds on agentic frameworks like ReAct [12] and Toolformer [13], which use tool calls to retrieve information. We specifically focus on optimizing this action policy inside resource-constrained edge-computing environments.

3 Methodology & Architecture

3.1 Model Quantization & Target Layer Selection

We load the base Qwen2.5-3B-Instruct model in 4-bit double quantization format using bitsandbytes. To optimize for Turing architecture (RTX 2080 Ti), we enforce bfloat16 compute and loading types, bypassing the PyTorch GradScaler which natively lacks support for scaling mixed-precision gradients and frequently crashes during causal language model training loops.

Unlike typical LoRA configurations which target attention projections (`q_proj`, `v_proj`), we explicitly target the MLP feed-forward blocks:

$$\text{Target Modules} = [\text{gate_proj}, \text{up_proj}, \text{down_proj}] \quad (1)$$

Mathematically, a standard SwiGLU MLP layer block (used in Qwen2.5 and Llama architectures) is represented as:

$$\text{MLP}(x) = W_{\text{down}} \left(\text{siLU}(W_{\text{gate}}x) \odot W_{\text{up}}x \right) \quad (2)$$

where $\odot$ denotes the element-wise Hadamard product, and $\text{siLU}(a) = a \cdot \sigma(a)$ is the Sigmoid Linear Unit.

For a quantized base layer weight $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, QLoRA decomposes the weight update ΔW via two low-rank matrices $A \in \mathbb{R}^{r \times d_{\text{in}}}$ and $B \in \mathbb{R}^{d_{\text{out}} \times r}$ as:

$$W' = \text{dequantize}(W_{\text{nf4}}) + \frac{\alpha}{r} BA \quad (3)$$

where $r \ll \min(d_{\text{in}}, d_{\text{out}})$ is the rank and α is a constant scaling hyperparameter. Applying this formulation to all three projections inside the MLP blocks forces behavioral adaptivity into the MLP’s non-linear feed-forward mapping, leaving the attention projection structures frozen.

3.2 Cognitive Heuristic Dataset Schema

To teach the model search heuristics, we utilize standard ChatML formats to model an “internal monologue” trace. The training pair does not simply ask for an answer; it asks the model to output a `Thought Process`: demonstrating *how* and *why* it is looking at a specific file, followed by a formatted `Next Action`:

```
<|im_start|>system
You are an agentic AI system. You possess advanced
cognitive heuristics to scan messy server contexts...
Always explain your detailed internal thought
process (Thought Process)... then output your
final Next Action in the format:
Next Action: read <file_name><|im_end|>
<|im_start|>user
Task: Find SQL Injection.
Directory:
- nginx.log
- auth.log
Previous Action: None<|im_end|>
<|im_start|>assistant
Thought Process: To identify a SQL injection,
I must examine the HTTP web server requests.
I will scan nginx.log first.
Next Action: read nginx.log<|im_end|>
```

3.3 Dynamic Active-Inference & Sleep-Cycle Loop

The system is governed by a dynamic interaction pipeline that interleaves active-inference with an SFT “sleep cycle” (micro-training). During active diagnostics, the agent’s interaction steps (planning state, file context, and selected actions) are recorded in an in-memory session buffer.

When the buffer size reaches a threshold $T = 5$, a sleep cycle is automatically triggered:

1. The active inference model graph is deleted and VRAM is cleared (`cuda.empty_cache()`).
2. The base model is loaded in 4-bit double quantization, and the MLP LoRA adapters are attached.
3. The SFT training session is executed on the combined dataset (baseline traces plus new buffer traces) using `paged_adamw_8bit` optimizer and gradient checkpointing.
4. The trained LoRA weights are merged back, the training model is unloaded, and the updated adapter is reloaded in evaluation mode for subsequent active inference.

This dynamic active-sleep cycle ensures rapid capability convergence without exceeding VRAM constraints.

During execution, instead of feeding all logs into the model, the active loop is initialized with only a 190-token **Directory Listing**. If the model outputs `Next Action: read nginx.log`, the python runtime intercepts the call, fetches *only* the contents of `nginx.log` from the disk, and appends it to the prompt. This keeps active context extremely small.

4 Experimental Evaluation

4.1 Computational Setup

All experiments were conducted on a commodity edge workstation:

- **GPU:** 1x NVIDIA GeForce RTX 2080 Ti (11GB VRAM, Turing architecture, Compute Capability 7.5)
- **CPU:** 24-core Intel Processor (64GB host system RAM)
- **Software Stack:** PyTorch 2.1.2, Transformers 4.39.0, PEFT 0.9.0, TRL 0.8.1 (utilizing SFTConfig and processing_class).

4.2 Phase 1: 1,200-Token Server Log Haystack Search

In Phase 1, we evaluated the raw base model and the fine-tuned MLP-LoRA model on a 1,200-token server log haystack. The task was to search through a moderately sized log sequence to identify a buried unauthorized SSH login attempt ('anomalous_hacker' originating from IP '192.168.99.199'). We evaluated both models on search accuracy, Time to First Token (TTFT), total generation time, and formatting stability.

4.3 Phase 2: Extreme 13,546-Token Diagnostics Task

In Phase 2, we tested the models on a highly complex security diagnostics scenario. The objective was to trace a SQL Injection exploit vector and subsequent root privilege escalation chain hidden across 5 logs (nginx.log, syslog, database.log, auth.log, cron.log) totaling **13,546 tokens**. We compared four distinct execution scenarios:

1. **Scenario A (Baseline 3B GPU Ingestion):** Brute-forcing the full concatenated 13.5k tokens in a single prompt on the 3B base model.
2. **Scenario B (Heuristic 3B Agent):** The fine-tuned MLP-LoRA 3B model executing dynamic search loops step-by-step on GPU.
3. **Scenario C (Base 7B Model CPU Ingestion):** Loading Qwen2.5-7B-Instruct in CPU RAM and brute-forcing the full prompt.
4. **Scenario D (Base 7B Model GPU Ingestion - Estimated):** Extrapolated metrics of a 7B model brute-forcing on a high-VRAM GPU (RTX 3090/A100).

5 Results & Discussion

5.1 Phase 1 Results: Baseline vs. MLP-LoRA

Both models successfully located the SSH anomaly in Phase 1 (100% accuracy). However, they exhibited massive differences in computation speed and formatting stability, as summarized in Table 1.

Table 1: Phase 1 Haystack Search Performance Summary

Metric	Baseline (No LoRA)	Heuristic (MLP-LoRA)
Search Accuracy	PASS (100%)	PASS (100%)
TTFT	3.0792 s	2.9641 s (4% faster)
Total Gen Time	15.4839 s	9.4665 s (39% faster)
Formatting Conformance	Poor (Drifting)	Perfect (Stable)
Output Verbosity	High (256 tokens)	Low (72 tokens)

Qualitatively, the baseline model failed to strictly follow the desired system format. It generated a wordy, repetitive numbered monologue and used non-target headers:

Thought Process:

1. I need to identify the username and IP address...
2. The log entry that stands out is...

...

Final Next Action:

Analyze the log entry "sshd[22415]: Accepted...

In contrast, our MLP-LoRA model generated a concise, direct thought process and conformed perfectly to the target action syntax:

Thought Process: The final sshd connection event shows a successful authentication from IP 192.168.99.199 with the username anomalous_hacker. I will search the logs for any subsequent failed authentication attempts from this IP to confirm.

Next Action: search ip 192.168.99.199 failed auth

This concise planning output prevented the model from wasting compute budget on redundant tokens, contributing to the **39% generation speedup**.

5.2 Phase 2 Results: Extreme Context Diagnostics

The performance summary across all scenarios is tabulated in Table 2.

5.3 Discussion of Computational Tradeoffs

The experimental results yield four critical takeaways:

1. **The Brute-Force Barrier:** Ingesting 13,500+ tokens on a consumer-grade 11GB VRAM card is a physical impossibility under standard execution (Scenario A). Standard prefill operations attempted to allocate 10.94 GiB of memory, immediately triggering a hardware CUDA OOM crash.
2. **The CPU Latency Penalty (720x Gap):** While Scenario C successfully bypasses physical VRAM limits by utilizing host system RAM, prefilling 13,546 tokens in float16 on the CPU takes **46.89 minutes (2,813.12s)**. The lack of hardware FlashAttention on the CPU and slower FP16 emulation render brute-force host execution practically unusable for real-time applications.

Table 2: Phase 2 Extreme Context Performance Summary

Metric	Scenario A: 3B GPU Brute	Scenario B: 3B Heuristic	Scenario C: 7B CPU Brute	Scenario D: 7B GPU Brute
Search Accuracy	FAIL (0% - Crashed)	PASS (100% success)	PASS (100% success)	PASS (100% success)
Peak Memory	> 11.0 GB VRAM (OOM)	< 1.5 GB VRAM	18.2 GB CPU RAM	~14.5 GB VRAM
TTFT	N/A	0.84s	~2750.00s	~0.35s
Total Duration	N/A (Crashed)	39.10 seconds	2,813.12 seconds	~ 4.50 seconds
Compute Device	GPU (RTX 2080 Ti)	GPU (RTX 2080 Ti)	Host CPU (24 Cores)	High-VRAM GPU*
Task Outcome	OOM CRASH	SUCCESS	SUCCESS (Ultra-Slow)	SUCCESS

*Note: Scenario D represents estimated performance on a high-VRAM GPU (e.g., 24GB RTX 3090/4090 or 80GB A100) running the 7B model in 4-bit with GPU FlashAttention.

3. **Heuristic Efficiency:** Scenario B solved the identical exploit mapping task on the same commodity GPU in **39.10 seconds**. By planning and loading files dynamically, the model kept its active context sequence under **2,741 tokens** at all times, achieving a **72x speedup** compared to CPU brute-force, while completely bypassing the OOM crash.

4. **FFN vs. Attention Adaptation:** Adjusting the FFN weights (via MLP-LoRA) effectively adapts the model’s action-selection policy (cognitive search heuristics) while leaving the base self-attention projection structures frozen. This decouples semantic feature extraction (attention heads) from behavioral execution (MLP blocks). We hypothesize that this decoupling is why MLP-LoRA maintains 100% reasoning accuracy and prevents formatting drift over extended dialog turns.

6 Conclusion & Future Work

In this paper, we proved that targeting Multilayer Perceptron (MLP) layers with QLoRA successfully trains language models to execute advanced cognitive search heuristics. In resource-constrained environments, teaching a smaller model *how* to plan and dynamically fetch context step-by-step is computationally superior to brute-force context window ingestion.

Our fine-tuned 3B model bypassed VRAM hardware ceilings, matched the accuracy of a model double its size (7B), and executed **72 times faster** than CPU brute-force. Future work will explore automating active-sleep cycle micro-training loops inside distributed edge-computing networks to enable continuous self-improvement of edge agents.

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